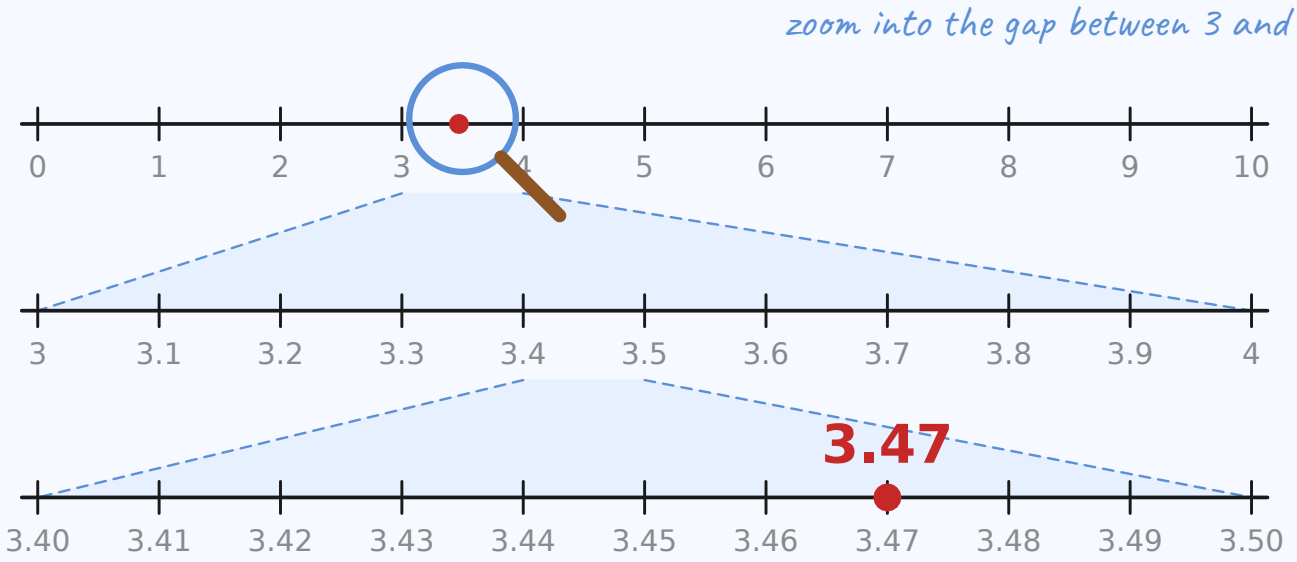
Chapter 3

Decimal fractions

numbers with a dot in them, seen through a magnifying glass



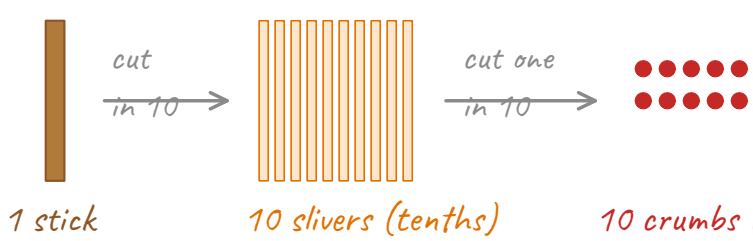
A whole number counts whole things. But life is full of bits: £3.47, 1.8 metres, 37.5 °C. The **dot** in these numbers lets you say "and a bit" precisely. This chapter shows that a decimal is nothing new: it's the same place-value idea from Chapter 1, carried on **to the right**, and it's another way to write a fraction (Chapter 2).

You'll learn to read, compare, add, multiply and divide decimals, turn fractions into decimals with a comic-strip division, and round numbers sensibly (decimal places and significant figures, both on your Lecture 1 slides).

Goes with **Lecture 1** (significant figures, decimal places, recurring decimals).

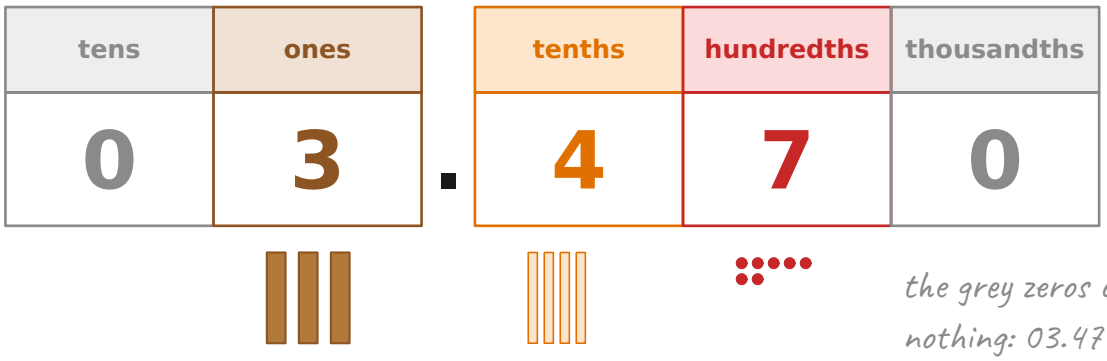
Cutting a stick into ten

In Chapter 1, ten sticks made a bundle. Now go the other way. Take **one stick** and cut it into **ten equal slivers**. Each sliver is a tenth of a stick. Cut one sliver into ten crumbs: each crumb is a hundredth of a stick.



1 stick = 10 slivers
1 sliver = 10 crumbs
1 stick = 100 crumbs

Now **3 sticks, 4 slivers and 7 crumbs** is written like this. The **dot** (called the **decimal point**) is a fence: whole sticks on the left, pieces on the right.



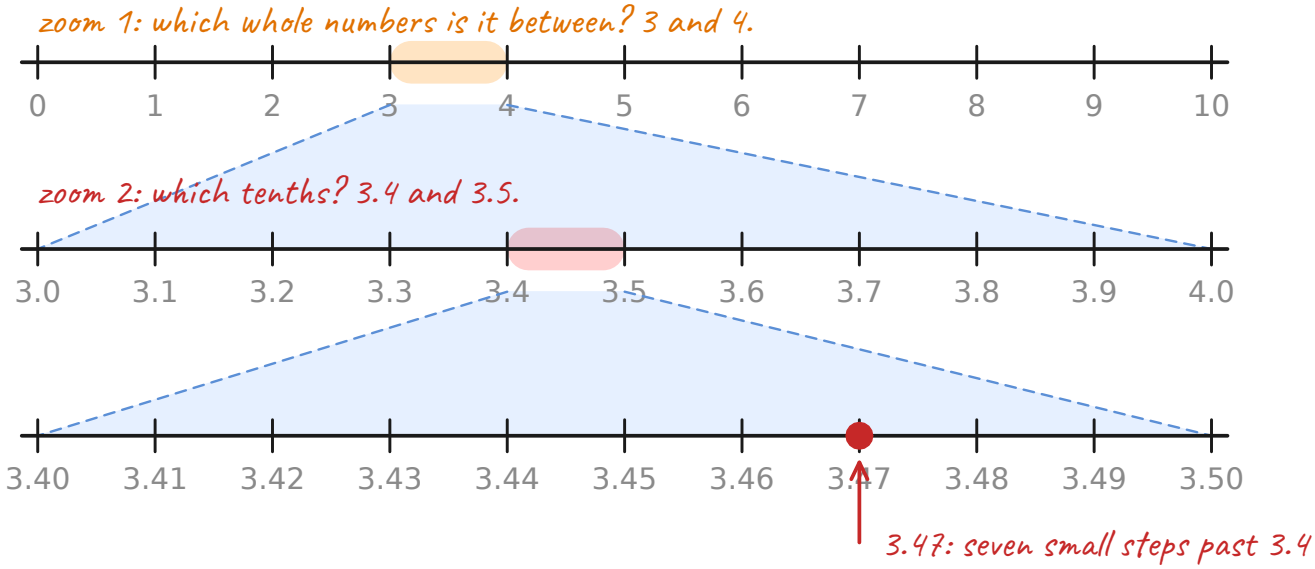
the grey zeros change nothing: 03.470 = 3.47

Read it as **"three point four seven"**. (Not "three point forty-seven": after the dot you read the digits one at a time.)

Each column is **ten times smaller** than the column to its left, on **both** sides of the dot. That's the only rule. Hundreds, tens, ones, tenths, hundredths, thousandths... one long ladder, and the dot just marks where the whole ones stop.

The magnifying glass: where does 3.47 live?

A number line is a ruler that goes on forever. Decimals live **in the gaps** between whole numbers. To find 3.47, zoom in twice:



Every zoom shows one more digit. The first digit after the dot says which tenth, the next says which hundredth, and so on. You could keep zooming forever, which is why some numbers need endless digits (you'll meet them later in this chapter).

Which is bigger: 0.5 or 0.45?

Lots of people say 0.45, because 45 is bigger than 5. **That's the trap.** Compare column by column, from the left, like a race where the first difference wins:



With slivers you can see it: half a stick (5 slivers) is more than 4 slivers and a few crumbs. **Line the dots up, pad with zeros, compare from the left.**

A decimal is a fraction wearing a disguise

PROCEDURE: decimal to fraction

1. Count the digits after the point: 1 digit means tenths, 2 hundredths, 3 thousandths.
2. Write the digits after the point over 10, 100 or 1000 (one 0 per digit), then simplify.
3. A whole-number part stays in front (mixed number): $2.6 = 2\frac{6}{10} = 2\frac{3}{5}$.

The last column you use tells you the bottom number. Read the digits after the dot as a whole number, and put it over 10, 100 or 1000:

$$0.7 = \frac{7}{10}$$ *one digit after the dot: tenths*

$$0.25 = \frac{25}{100} = \frac{1}{4}$$ *two digits: hundredths, then simplify*

$$0.125 = \frac{125}{1000} = \frac{1}{8}$$ *three digits: thousandths*

$$2.6 = 2\frac{6}{10} = 2\frac{3}{5}$$ *the whole 2 just rides along*

Worth knowing by heart (they come up constantly):

$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{8}$	$\frac{1}{3}$
0.5	0.25	0.75	0.2	0.1	0.125	$0.\dot{3}$

(The dot over the 3 in the last one is explained later in this chapter.)

🔍 Adding and taking away: line up the dots

PROCEDURE: adding and subtracting decimals

1. Write the numbers in columns with the **decimal points in a vertical line**.
2. Fill empty places with zeros so every number has the same number of decimal places ($5 = 5.00$).
3. Add or subtract column by column from the right, exactly as for whole numbers.
4. The point in the answer goes directly under the other points.

A shopping receipt already does this. Pence line up under pence, pounds under pounds, so the dots form a straight vertical line:

CORNER SHOP

milk	1.35
bread	0.99
cheese	12.50
TOTAL	14.84

the dots sit in one straight column.

Then add exactly like Chapter 1:

hundredths: 5 + 9 + 0 = 14, write 4 carry 1

tenths: 3 + 9 + 5 + 1 = 18, write 8 carry 1

ones: 1 + 0 + 2 + 1 = 4

tens: 1

the dot drops straight down into the answer

Taking away: fill the empty spaces with zeros first. Work out $5 - 1.37$:

$$\begin{array}{r} 5.00 \\ -\ 1.37 \\ \hline 3.63 \end{array}$$

5 is the same as 5.00 (5 pounds = 5 pounds 0 pence).

Write the zeros in so every column has a digit.

then take away exactly as in Chapter 1, opening bundles when a column is too small.

check: 3.63 + 1.37 = 5.00 ✓

Times 10: the dot stays still, the digits slide

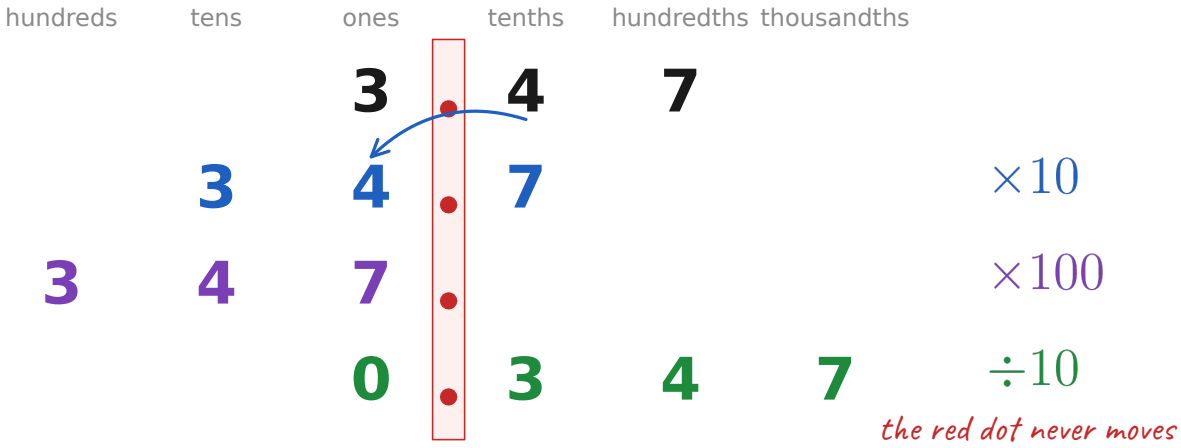
RULE: multiplying and dividing by 10, 100, 1000

$\times 10$, $\times 100$, $\times 1000$: every digit moves **1, 2 or 3 places left** (the number gets bigger). $3.47 \times 100 = 347$.

$\div 10$, $\div 100$, $\div 1000$: every digit moves **1, 2 or 3 places right** (the number gets smaller). $3.47 \div 10 = 0.347$.

Fill any gap next to the point with 0: $0.6 \div 100 = 0.006$.

Multiplying by 10 makes every stick into a bundle, every sliver into a stick, every crumb into a sliver. Each digit moves **one column to the left**. Picture the dot nailed to the page and the digits sliding past it:

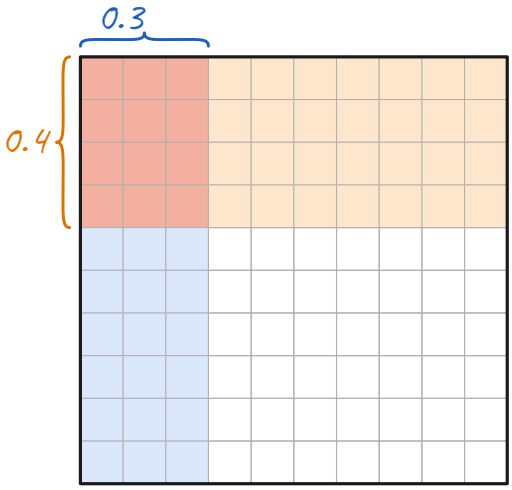


× 10: digits slide 1 place left. **× 100**: 2 places. **× 1000**: 3 places. **÷ 10**: 1 place right, and so on. Gaps next to the dot get filled with zero (0.347 needs a 0 in the ones column so the dot isn't left floating).

Shortcut people say: "move the decimal point". It gives the same answer, but it's really the digits moving; the columns stay where they are.

🔍 Multiplying decimals: a square cut into 100

What is 0.3×0.4? Think of "of": 0.3 **of** 0.4. Take a square that is 1 by 1, cut into a 10 × 10 grid. Shade 3 columns (0.3 of the width) and 4 rows (0.4 of the height):



The overlap (red) is 3 × 4 = 12 little squares.

Each little square is 1/100 of the whole.

$$0.3 \times 0.4 = \frac{12}{100} = 0.12$$

the answer is SMALLER than both. Multiplying by something less than 1 shrinks, like taking a part of a part.

PROCEDURE: multiplying decimals

1. Ignore the decimal points and multiply the whole numbers.
2. Count the digits after the point in **both** numbers together.
3. Give the answer that many digits after the point (count from the right; add 0s at the front if needed: $0.03 \times 0.2 = 0.006$).
4. **Then** you may drop zeros at the end: 0.5×0.4: $5 \times 4 = 20$, 2 places, $0.20 = 0.2$.

2.5×1.3 ← *1. ignore the dots for a moment*

$25 \times 13 = 325$ ← *2. multiply whole numbers (Chapter 1 grid method)*

$2.5 \rightarrow 1$ digit, $1.3 \rightarrow 1$ digit ← *3. count digits after the dots: 1 + 1 = 2*

3.25 ← *4. give the answer 2 digits after the dot*

$0.5 \times 0.4 = 0.20 = 0.2$ ← *5 × 4 = 20; 2 places gives 0.20; the end 0 can go*

Place the point before you drop the zero. Write 20, count 2 places from the right: 0.20. Dropping the 0 first and then counting 2 places gives the wrong 0.02.

Sanity check: 2.5 × 1.3 is roughly 2.5 × 1 = 2.5, a bit more. 3.25 is sensible; 32.5 or 0.325 would be silly. Always ask "is the answer the right size?"

Dividing by a decimal: make it a whole number first

How many 15p coins fit into £4.50? That's $4.5 \div 0.15$. Dividing by 0.15 feels impossible, but in **pence** the question is easy: how many 15s in 450?



... 30 coins make 450p

$$\frac{4.5}{0.15} = \frac{4.5 \times 100}{0.15 \times 100} = \frac{450}{15} = 30$$

multiply top and bottom by the same number: that's an equivalent fraction (Chapter 2), so the answer can't change

PROCEDURE: dividing by a decimal

1. Multiply **both** numbers by the same power of 10 (10, 100, ...) so that the number you divide **by** becomes a whole number.
2. Divide normally (short division). The answer is unchanged, because $\frac{a}{b} = \frac{10a}{10b}$.
3. Check by multiplying back: answer × divisor = the number you started with.

$$0.36 \div 0.4 = \frac{3.6}{4} = 0.9$$ ← *× 10 top and bottom; check 0.9 × 0.4 = 0.36*

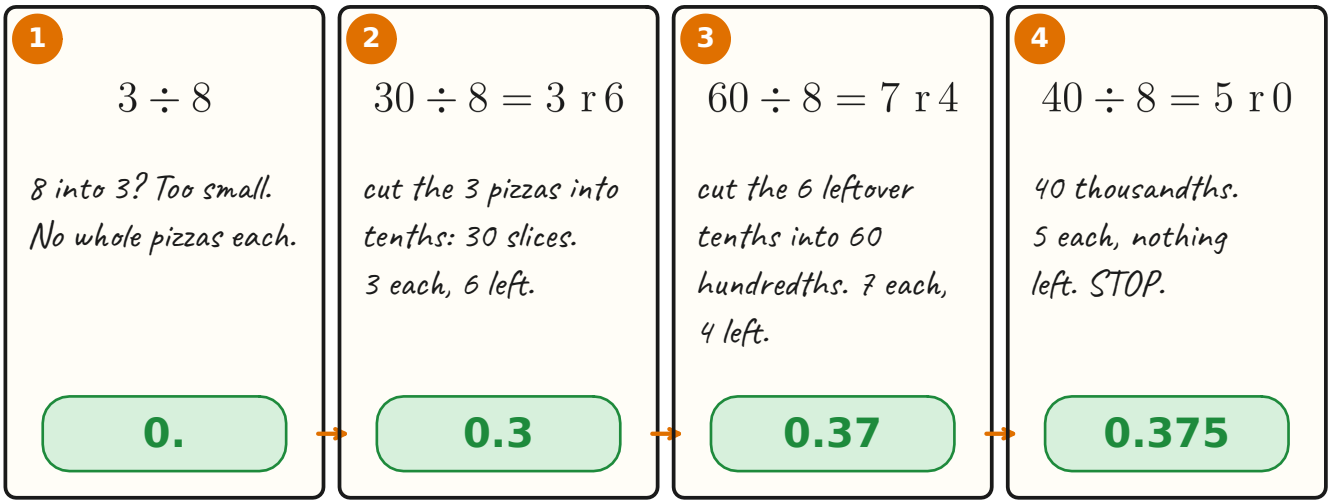
Dividing by a number **less than 1** makes the answer **bigger** (lots of small coins fit). That's the mirror of multiplying on the previous page.

🔍 Fraction to decimal: division as a comic strip

PROCEDURE: fraction to decimal

1. $\frac{a}{b} = a \div b$: divide the top by the bottom.
2. Write the top as $a.000\ldots$ and keep dividing; each remainder carries to the next column.
3. Stop when the remainder is 0 (**terminating** decimal) or a remainder repeats (**recurring** decimal: the digits repeat from there on).

A fraction is a division (Chapter 2): $\frac{3}{8}$ means $3 \div 8$. Eight people share 3 pizzas. Follow the strip one panel at a time:



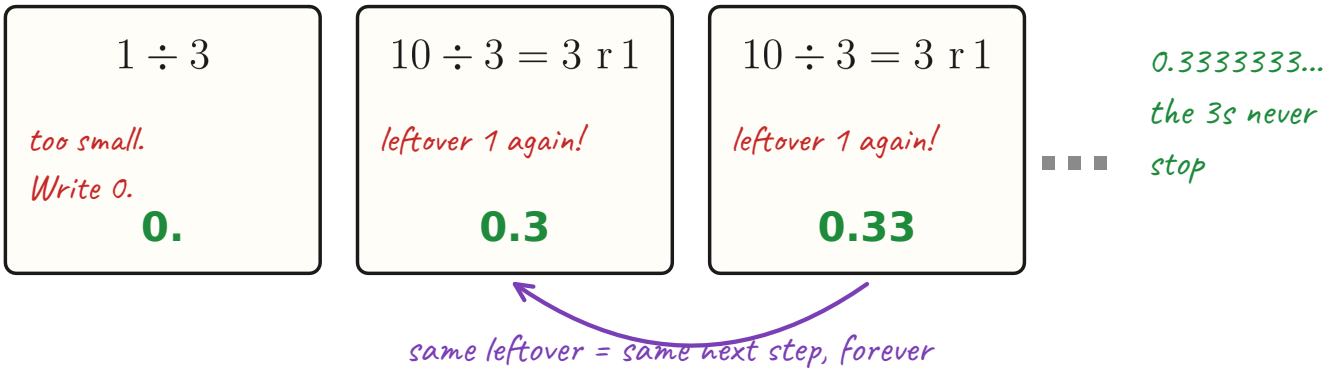
Written the usual compact way ("bus stop" division), the same four panels look like this. The small red numbers are the leftovers carried to the next column:

$$
\begin{array}{r|l}
 & 0.3\ 7\ 5 \\
\hline
8 & 3.{}^{3}0\,{}^{6}0\,{}^{4}0
\end{array}
$$

the carried 3, 6 and 4 are the leftovers from panels 1, 2 and 3. Add as many 0s after the dot as you need: 3 = 3.000

When the strip never ends: recurring decimals

Try $\frac{1}{3} = 1 \div 3$. Panel after panel, the same leftover comes back:



A decimal that repeats forever is called **recurring**. We can't write infinitely many 3s, so we put a **dot over the repeating digit**:

$\frac{1}{3} = 0.\dot{3}$ *0.3333...*

$\frac{1}{6} = 0.1\dot{6}$ *only the 6 repeats: 0.16666...*

$\frac{2}{11} = 0.\dot{1}\dot{8}$ *dots on first and last of the repeating block: 0.181818...*

$\frac{1}{7} = 0.\dot{1}42857\dot{}$ *block 142857 repeats*

Your Lecture 1 slide writes $\frac{1}{3} = 0.3\dot{3} \approx 0.33$. The wavy equals sign $\approx$ means **"is roughly"**: 0.33 is close to a third, but not exactly it.

Why the lecture says keep fractions when you want exact answers: $\frac{1}{3} \times 3 = 1$ exactly, but $0.33 \times 3 = 0.99$. The rounding error sneaks in. Use fractions for algebra, decimals when an approximate answer is fine (statistics, measurements).

Which fractions stop, and recurring back to fractions

RULE: terminating or recurring?

Simplify the fraction first.

If the denominator's prime factors are **only 2s and 5s**, the decimal **terminates** (stops). Reason: it can be scaled to a denominator of 10, 100, 1000, ... (and $10 = 2 \times 5$).

Any other prime factor (3, 7, 11, ...) in the denominator gives a **recurring** decimal.

Fraction	Denominator	Primes	Decimal
$\frac{3}{8}$	8	2^3	**terminates**: 0.375
$\frac{7}{20}$	20	$2^2 \times 5$	**terminates**: 0.35
$\frac{5}{12}$	12	$2^2 \times 3$	**recurs**: $0.41\dot{6}$
$\frac{9}{15}$	$15 \to 5$ after simplifying to $\frac{3}{5}$	5	**terminates**: 0.6

PROCEDURE: recurring decimal to fraction

1. Call the decimal x. Write out several repeats: $x = 0.272727\ldots$
2. Multiply by 10 if one digit repeats, 100 if two digits repeat, 1000 if three, ... so the repeating part lines up: $100x = 27.2727\ldots$
3. Subtract: the endless tails cancel. $100x - x = 27$, so $99x = 27$.
4. Divide and simplify: $x = \frac{27}{99} = \frac{3}{11}$.

Example. Write $0.\dot{3}$ as a fraction.

$$x = 0.3333\ldots$$ ← one digit repeats: use 10x

$$10x = 3.3333\ldots$$ ← × 10

$$10x - x = 3 \;\Rightarrow\; 9x = 3$$ ← the tails .3333... cancel

$$x = \frac{3}{9} = \frac{1}{3}$$ ← ÷ 9, simplify

Example. $0.1\dot{6}$ (only the 6 repeats): use $10x$ and $100x$, which both end in $.666\ldots$

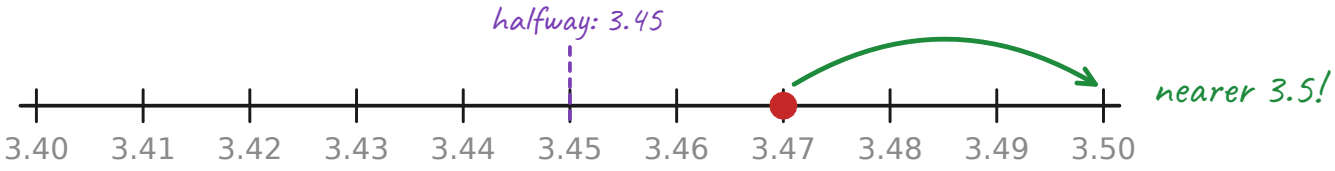
$$100x = 16.666\ldots, \quad 10x = 1.666\ldots$$

both tails are .666...

$$90x = 15 \;\Rightarrow\; x = \frac{15}{90} = \frac{1}{6}$$ ← *subtract, then ÷ 90*

Rounding: which mark on the ruler is nearer?

A price of £3.47 on a sign that only shows 10p steps: is it nearer £3.40 or £3.50? Zoom in with the magnifying glass:



So 3.47 **rounded to 1 decimal place** is 3.5. "Decimal places" (d.p.) just means digits after the dot. You don't have to draw a ruler every time. The rule is:

PROCEDURE: rounding to a number of decimal places

1. Find the last digit you are keeping (the **cut**).
2. Look at the **next** digit only (the **decider**).
3. Decider **5 or more**: increase the kept digit by 1. Decider **4 or less**: leave it.
4. Remove every digit after the cut. Keep end zeros that are inside the required places (3.96 to 1 d.p. is 4.0, not 4).

Round 2.71828 to 2 decimal places:

2. 71|**8**28 → **2.72**

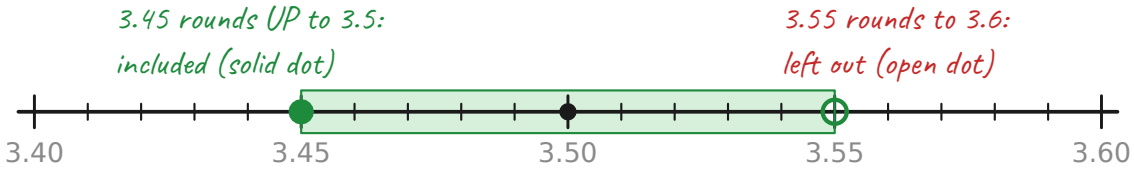
cut *decider 8*

8 is 5 or more: the 1 bumps up to 2

Bumping can ripple: 3.96 to 1 d.p. The decider 6 bumps the 9 up, 9 + 1 = 10, so it carries: 3.96 becomes **4.0**. Keep the 0: it shows you rounded to 1 d.p.

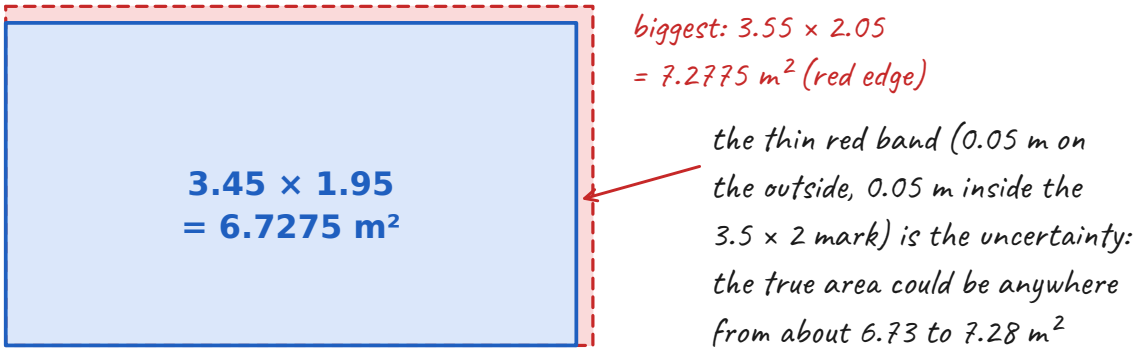
Going backwards: what could it have been? A tape measure says a rug is **3.5 m** long, rounded to 1 d.p. The true length isn't exactly 3.5.

> ✎ **YOUR GO (pencil first)**
>
> *On a ruler from 3.4 to 3.6, shade every true length that would round to 3.5. Where does the shading start, and where does it stop?*



So "3.5 to 1 d.p." really means **anywhere from 3.45 up to (but not including) 3.55**: half a step either side of the mark. Those two ends are the **lower and upper bounds**.

Why it matters: the rug's area. It's 3.5 m by 2.0 m, both to 1 d.p., so "7 m²". But the true rug is somewhere between the smallest and the biggest possible rectangles:



A thin half-step on each side becomes a whole band of area: about 0.28 m² either way. Multiplying rounded numbers multiplies their uncertainty too, so don't quote "7.000 m²" from measurements that were only good to 1 d.p.

🔍 Upper and lower bounds: the rules

PROCEDURE: bounds of a rounded number

1. Find the step size of the rounding: 1 d.p. means steps of 0.1, 2 d.p. steps of 0.01, nearest 10 means steps of 10.
2. **Half a step** either side: lower bound $=$ value $-$ $\frac{1}{2}$ step, upper bound $=$ value $+$ $\frac{1}{2}$ step.
3. Write $\text{lower} \le x < \text{upper}$: the lower bound is included, the upper bound is not (it would round up).

$$7.8 \text{ (1 d.p.)}: \quad 7.75 \le x < 7.85$$ ← step 0.1, half step 0.05

$$70 \text{ (nearest 10)}: \quad 65 \le x < 75$$ ← step 10, half step 5

RULE: bounds of a calculation

Add: biggest $=$ upper $+$ upper; smallest $=$ lower $+$ lower. Same idea for $\times$.

Subtract: biggest $=$ upper $-$ **lower**; smallest $=$ lower $-$ **upper**.

Divide: biggest $=$ upper $\div$ **lower**; smallest $=$ lower $\div$ **upper**. (A big top over a small bottom gives the biggest answer.)

Example (division). A plank is 2.4 m long and is cut into pieces 0.3 m long, both measured to 1 d.p. Find the bounds for the number of pieces, $2.4 \div 0.3$.

$$2.35 \le \text{plank} < 2.45, \qquad 0.25 \le \text{piece} < 0.35$$

half a step (0.05) each side

$$\text{max} = \frac{2.45}{0.25} = 9.8$$ ← upper ÷ lower

$$\text{min} = \frac{2.35}{0.35} \approx 6.71$$ ← lower ÷ upper

So "8 pieces" ($2.4 \div 0.3$) could really be anything from about 6.7 to 9.8: rounding in a division can matter a lot.

🔍 Rounding versus chopping (truncating)

Your Lecture 1 slide says 0.123456789 "truncated" to 2 d.p. is 0.12. To **truncate** means to just **chop off** the extra digits without looking at the decider:

0.678 *round* ⟶ **0.68** *decider 8 → bump the 7 up*

0.678 *truncate* ⟶ **0.67** *just chop after 2 digits*

For 0.123456789 both give 0.12 (the decider is 3, so nothing bumps). They only differ when the decider is 5 or more. Unless told to truncate, **round**.

Significant figures: a rough ruler and a fine ruler

RULE: which digits are significant?

Non-zero digits always count. 4.72 has 3 s.f.

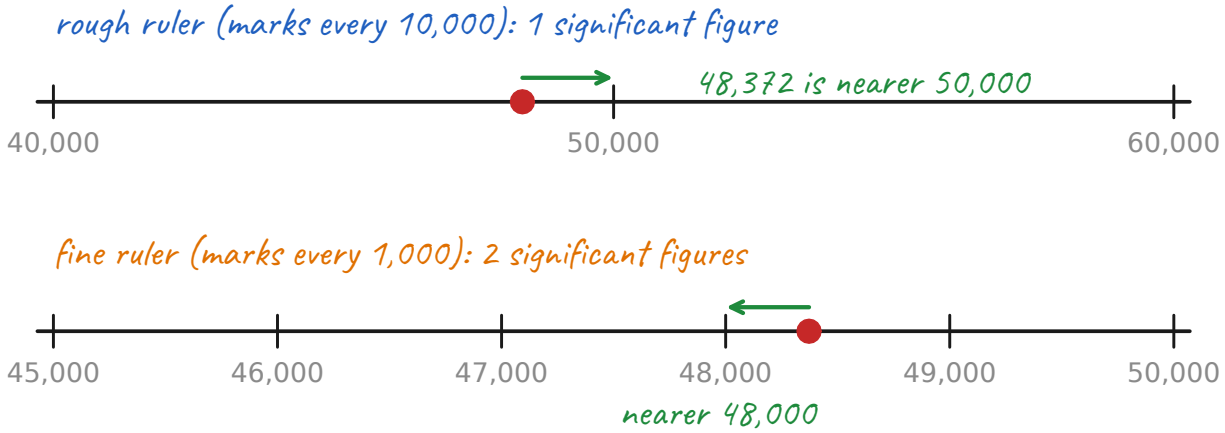
Leading zeros never count: zeros before the first non-zero digit only hold place. 0.0047 has 2 s.f. (the 4 and the 7).

Zeros between non-zero digits always count: 3.07 has 3 s.f.; 2005 has 4 s.f.

Trailing zeros after the decimal point count: they were written on purpose to show precision. 0.030 has 2 s.f. (the 3 and the final 0); 4.0 has 2 s.f.

Trailing zeros in a whole number are ambiguous: $48\,000$ could be 2, 3, 4 or 5 s.f. Say which ("48 000 to 2 s.f.") when it matters.

A crowd of **48,372** people. A newspaper says "about 50,000"; a report says "48,000". Both are rounding, just with rulers of different fineness:



Significant figures (s.f.) count digits **from the first non-zero digit**. They say how fine your ruler is, whatever the size of the number. Zeros at the end of a big number stay, because they hold the size (48,000 must not shrink to 48).

Small numbers: the zeros at the front are only place-holders, so they don't count. Your Lecture 1 table, with the counting shown:

0. 123456789

1 2 3 4 5 6 7 8 9

leading 0s: not counted

1 s.f.	**0.1**
2 s.f.	**0.12**
3 s.f.	**0.123**
4 s.f.	**0.1235**

4 s.f.: decider 5, so 4 bumps to 5!

Heads-up: the slide table shows 0.1234 for 4 s.f.; that is the **truncated** value. Rounded properly it's 0.1235, because the next digit is 5.

One more: 0.004 72 to 2 s.f. Skip the zeros, the first significant digit is 4. Keep 4 and 7, decider 2: **0.0047**.

PROCEDURE: rounding to n significant figures

1. Start counting at the **first non-zero digit**. Count n digits (zeros in between count). That is the cut.
2. Round at the cut using the decider, as for decimal places.
3. **Whole-number part:** replace the dropped digits with zeros to keep the size ($48\,372 \to 48\,000$). **After the point:** just drop them.

Number	s.f.	Why
3.07	3	the 0 is between 3 and 7: counts
0.030	2	leading zeros don't count; the final 0 after the point does
0.0405	3	4, 0, 5 (the zero between counts)
12.50	4	trailing zero after the point counts
5200	2, 3 or 4	trailing zeros in a whole number: ambiguous

$3.0496 \to 3.05$ (3 s.f.) ← *cut after 3, 0, 4; decider 9: 4 → 5*

$2045 \to 2000$ (2 s.f.) ← *keep 2, 0; decider 4; refill with zeros*

$0.06996 \to 0.0700$ (3 s.f.) ← *6, 9, 9; decider 6 ripples: keep both zeros*

Estimating: round everything to 1 s.f. to check

Before trusting a calculator, round each number to 1 significant figure and do the easy sum in your head. If the real answer is nowhere near, you pressed a wrong button.

$$4.87 \times 21.3$$ ← the real question

$$\approx 5 \times 20$$ ← 1 s.f. each: 4.87 → 5, 21.3 → 20

$$= 100$$ ← easy!

The calculator says 103.731. Close to 100 ✓. If it said 1037.31 you'd know a dot went astray.

Find the unknown

PROCEDURE: finding an unknown decimal

1. Undo the operation with its inverse ($\times$ with $\div$, $+$ with $-$), then use the decimal rules above.
2. For "it was rounded to ..." questions, the unknown is a **range**: give the lower and upper bounds.
3. Check by putting the answer back.

Example. $? \times 0.4 = 0.36$

$$? = 0.36 \div 0.4 = 3.6 \div 4 = 0.9$$ ← undo × 0.4; × 10 top and bottom

$$0.9 \times 0.4 = 0.36 \checkmark$$ ← 9 × 4 = 36, 2 places

Example. $? + 2.75 = 5.1$

$$? = 5.10 - 2.75 = 2.35$$ ← undo + with −; pad 5.1 to 5.10

Example. A length rounded to 1 d.p. is 2.7 cm. What could it really be?

$$2.7 - 0.05 = 2.65, \quad 2.7 + 0.05 = 2.75$$

half of the 0.1 step each side

$$2.65 \le x < 2.75$$ ← 2.75 itself would round to 2.8

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Place value	each column is $\frac{1}{10}$ of the one to its left: tenths, hundredths, thousandths
Compare	line up the points, pad with zeros, compare from the left
Decimal to fraction	$0.abc = \dfrac{abc}{1000}$ one 0 per decimal place, then simplify
Fraction to decimal	$\dfrac{a}{b} = a \div b$
+ and −	line up the points; fill gaps with 0
× and ÷ by 10, 100, 1000	digits move 1, 2, 3 places (left for ×, right for ÷)
Multiply	multiply without points; answer has as many decimal places as both numbers together
Divide	$\dfrac{a}{b} = \dfrac{10a}{10b}$ scale both until the divisor is whole
Terminating?	simplified denominator has only the primes 2 and 5
Recurring to fraction	$x = 0.\dot{2}\dot{7} :\ \ 100x - x = 27 \Rightarrow x = \dfrac{27}{99}$ multiply by 10, 100, ... (one 0 per repeating digit), subtract
Rounding	decider 5 or more: round up; 4 or less: leave
Significant figures	count from the first non-zero digit; zeros between count; trailing zeros after the point count; leading zeros never
Bounds	$\text{value} - \frac{1}{2}\text{step} \le x < \text{value} + \frac{1}{2}\text{step}$
Bounds of a ÷	$\max \dfrac{a}{b} = \dfrac{a_{\max}}{b_{\min}}, \quad \min \dfrac{a}{b} = \dfrac{a_{\min}}{b_{\max}}$
Estimate	round every number to 1 s.f. first

🔍 Magnifying glass: your turn

Answers are at the bottom of each group, small and grey. Cover them first!

Reading and comparing

1. What is the 6 worth in 2.068?
2. Which is bigger: 0.7 or 0.69?
3. Put in order: 0.3, 0.33, 0.303
4. Write 0.35 as a fraction (simplest)

1. 6 hundredths 2. 0.7 3. 0.3, 0.303, 0.33 4. 7/20

Working with them

5. 2.4 + 13.75
6. 10 - 3.62
7. 0.6 × 0.7
8. 1.2 × 3.5
9. 7.2 ÷ 0.08

5. 16.15 6. 6.38 7. 0.42 8. 4.2 9. 90

Fraction to decimal

10. 5/8
11. 2/3
12. 7/20

10. 0.625 11. 0.6 recurring 12. 0.35

Rounding

13. 6.748 to 1 d.p.
14. 6.748 to 2 d.p.
15. 0.03049 to 2 s.f.
16. 72,650 to 2 s.f.
17. Estimate 9.8 × 51

13. 6.7 14. 6.75 15. 0.030 16. 73,000 17. 10 × 50 = 500

Significant figures, recurring, bounds

18. How many s.f. in 3.07? In 0.030? In 0.0405?
19. Round 0.04096 to 3 s.f.
20. Write 0.4 recurring (0.444...) as a fraction
21. Does 7/40 terminate? Does 5/6?
22. Bounds of 12.6 (rounded to 1 d.p.)
23. a = 8 and b = 2, both to the nearest whole number. Largest a ÷ b?

18. 3, 2, 3 19. 0.0410 (4, 0, 9; decider 6 bumps the 9) 20. 4/9 ($10x - x = 4$) 21. yes ($40 = 2^3 \times 5$); no (6 has a 3) 22. $12.55 \le x < 12.65$ 23. 8.5 ÷ 1.5 = 5.67 (upper ÷ lower)

Find the unknown

24. ? × 0.4 = 0.36
25. ? − 1.8 = 2.65
26. ? ÷ 0.2 = 15
27. A number rounded to 1 d.p. is 4.3. What are its bounds?
28. A number rounded to 2 s.f. is 0.56. What are its bounds?

24. 0.9 (0.36 ÷ 0.4 = 3.6 ÷ 4) 25. 4.45 (2.65 + 1.8) 26. 3 (15 × 0.2) 27. $4.25 \le x < 4.35$ (± 0.05) 28. $0.555 \le x < 0.565$ (± 0.005)